```
Raw Data (CSV / Telco Churn Dataset)
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        ▼
┌──────────────────────────────────────────────┐
|  data_loader.py                              |
| ──────────────────────────────────────────── |
|  • Load dataset (pandas.read_csv)            |
|  • Data quality validation                   |
|    (missing values, duplicates, schema)      |
|  • Initial preprocessing                     |
|    (cleaning, type casting, basic imputation)|
└──────────────────────────────────────────────┘
        |
        ▼
┌──────────────────────────────────────────────┐
|  features.py                                 |
| ──────────────────────────────────────────── |
|  • Feature engineering & transformation      |
|  • Categorical encoding                      |
|    (label / one-hot / target encoding)       |
|  • Binary / derived features                 |
|    (tenure groups, charges ratios, etc.)     |
|  • Scaling / normalization (if needed)       |
|  • Save processed features (parquet / csv)   |
```

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└──────────────────────────────────────────┘
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        ▼
┌──────────────────────────────────────────┐
| models.py                                |
| ──────────────────────────────────────── |
| • Model training (XGBoost / LightGBM / etc.)|
| • Hyperparameter tuning (Optuna)         |
| • Cross-validation & evaluation          |
|   (accuracy, precision, recall, AUC, etc.) |
| • Log everything to MLflow               |
|   (params, metrics, tags, artifacts)     |
| • Register best model in MLflow Model    |
|   Registry                               |
└──────────────────────────────────────────┘
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        ▼
┌──────────────────────────────────────────┐
| tests/                                   |
| ──────────────────────────────────────── |
| • Unit tests (pytest)                    |
|   - Test data loading & validation       |
|   - Test feature transformations         |
|   - Test model prediction consistency    |
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│  • Sanity / smoke checks                  │
│  • Integration tests (end-to-end subset)    │
└──────────────────────────────────────────────┘
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┌──────────────────────────────────────────────┐
│  run_pipeline.py                            │
│  ──────────────────────────────────────────  │
│  • Orchestrate full end-to-end workflow     │
│    (data → features → train → evaluate)     │
│  • Configurable via YAML / CLI args         │
│  • Trigger MLflow run                       │
│  • Log full pipeline artifacts              │
│  • Optional: conditional retraining logic   │
└──────────────────────────────────────────────┘
        │
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┌──────────────────────────────────────────────┐
┌──────────────────────────────┐
│          MLflow Tracking Server          │◄──┤  MLflow Artifacts / Models    │
│  • Experiments, runs, metrics, params      │   │  • Stored models (.pkl / .onnx)│
│  • UI for comparison & model selection     │   │  • Preprocessor artifacts      │
│  • Model Registry (staging → production)    │
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┌──────────────────────────────────────────┐
│ Serving Layer (app/main.py)              │
│ ──────────────────────────────────────── │
│ • FastAPI application                    │
│ • Endpoints:                             │
│   - /predict (single / batch inference)  │
│   - /health                              │
│ • Load registered model from MLflow      │
│ • Preprocess input → predict → postprocess │
│ • Optional: Gradio / Streamlit UI wrapper  │
└──────────────────────────────────────────┘
        |
        ▼
┌──────────────────────────────────────────┐
│ Dockerization                            │
│ ──────────────────────────────────────── │
│ • Dockerfile (multi-stage recommended)   │
│ • Bundle: FastAPI + uvicorn + dependencies │
│ • Copy model artifacts / preprocessor    │
│ • Expose port 8000                       │
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┌──────────────────────────────────────────┐
│ CI/CD (GitHub Actions)                   │
│ ──────────────────────────────────────── │
│ • Lint, test, build Docker image         │
│ • Smoke / integration tests              │
│ • Push to Docker Hub / AWS ECR           │
│ • Optional: auto-deploy trigger          │
└──────────────────────────────────────────┘
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        ▼
┌──────────────────────────────────────────┐
│ AWS Deployment & Monitoring              │
│ ──────────────────────────────────────── │
│ • ECR → store Docker image               │
│ • ECS (Fargate preferred) / EC2          │
│ • Application Load Balancer (ALB)        │
│ • CloudWatch → logs + metrics            │
│ • Optional: auto-scaling, alarms         │
│ • End users → access via HTTPS endpoint  │
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```